Installation & Execution

Prerequisites

- Python 3.8 or higher
- Jupyter Notebook or Google Colab environment

Dependencies

Install the required libraries using below pip command:

```
pip install pandas numpy scikit-learn matplotlib seaborn nltk scipy
```

Setup Instructions

1. Clone the repository and open the notebook file (Zomato_Restaurant_Cluster_ML_Final_Submission.ipynb).

2. The notebook includes inline commands to download necessary NLTK datasets (stopwords, punkt_tab, wordnet). Ensure you have an active internet connection for this step.

3. The dataset links are hardcoded in the data loading section to fetch directly from cloud storage. No local data downloading is required.

4. The final execution block will generate the serialized model files (kmeans_model.pkl, hierarchical_model.pkl, std_scaler.pkl, pca_model.pkl, tfidf_vectorizer.pkl, variance_selector.pkl) in the working directory. These files contain the frozen pipeline required for the deployment phase.